

Troubleshooting Guide

Common Issues

1. UE5 Build Issues

“No GPU detected”

- **Cause:** UE5 requires a GPU for rendering and Pixel Streaming
- **Fix:** Use a GPU-enabled machine (AWS g4dn/g5, GCP N1+T4, Azure NV series)
- **Workaround:** Build scripts and asset prep can run on CPU-only machines

“GitHub 403 when cloning UE5”

- **Cause:** GitHub account not linked to Epic Games
- **Fix:**
 1. Go to <https://www.unrealengine.com/account/connections>
 2. Link your GitHub account
 3. Accept the Epic Games organization invite
 4. Use a Personal Access Token:

```
git clone https://<PAT>@github.com/EpicGames/UnrealEngine.git
```

Build fails with memory errors

- **Fix:** Ensure 32GB+ RAM; use `-MaxParallelActions 4` flag to reduce parallelism
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2. Streaming Issues

WebSocket connection refused

```
# Check if signalling server is running
curl http://localhost:8888/healthz

# If not running:
cd streaming/signalling && npm start

# Check firewall
sudo ufw allow 8888
sudo ufw allow 443
```

Video stream not appearing

- Ensure UE5 server is running with `-PixelStreamingURL=ws://signalling:8888`
- Check browser console for WebRTC errors
- Verify TURN server is accessible: `turnutils_uclient turn.finalevolutiongroup.com`

High latency (>100ms)

- Move GPU server closer to users (same region)
- Check network congestion: `mtr <server_ip>`
- Reduce streaming resolution in UE5 config
- Enable hardware encoding (NVENC)

Black screen on mobile

- Check if H.264 is supported (iOS requires H.264)
 - Verify WebRTC codec negotiation in browser console
 - Test with VP8 fallback
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3. iOS App Issues

Xcode signing errors

```
# Verify provisioning profile
security find-identity -v -p codesigning

# Clean build folder
xcodebuild clean -workspace FinalEvolutionLab.xcworkspace -scheme FinalEvolutionLab
```

App crashes on launch

- Check `Config.swift` URLs are correct
- Verify minimum iOS version is 16.0
- Check for missing privacy descriptions in `Info.plist`

Cannot connect to streaming server

- Verify ATS exceptions in `Info.plist`
 - Check if `stream.finalevolutiongroup.com` resolves on device
 - Test WebSocket connection: open Safari → `wss://stream.finalevolutiongroup.com`
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4. DNS Issues

Domain not resolving

```
# Check propagation
dig @8.8.8.8 finalevolutiongroup.com
dig @1.1.1.1 stream.finalevolutiongroup.com

# Check nameservers
dig NS finalevolutiongroup.com

# Flush local DNS cache
sudo systemd-resolve --flush-caches # Linux
sudo dscacheutil -flushcache        # macOS
```

SSL certificate errors

- Verify cert covers all subdomains
 - Check Cloudflare SSL mode matches server config
 - Ensure stream/turn subdomains are not proxied through Cloudflare
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5. Docker Issues

Container won't start

```
# Check logs
docker-compose logs -f signalling
docker-compose logs -f coturn

# Rebuild
docker-compose build --no-cache
docker-compose up -d
```

Port conflicts

```
# Find process using port
lsof -i :8888
lsof -i :3478

# Kill process
kill $(lsof -ti:8888)
```

6. Asset Pipeline Issues

DeepMotion API errors

- Verify `DEEPMOTION_CLIENT_ID` and `DEEPMOTION_CLIENT_SECRET` in `.env`
- Check API quota at <https://www.deepmotion.com/dashboard>
- Run `python scripts/process_elijahbonds_animations.py --test-auth`

Meshy generation failures

- Verify `MESHY_API_KEY` in `.env`
- Check task status: `python -c "from scripts.ai_asset_pipeline.meshy_service import MeshyService; ..."`
- API rate limit: wait 60s between requests

Missing assets in UE5

- Run `python scripts/ue5_setup/fel_verify_assets_standalone.py`
 - Check `GeneratedAssets/pipeline_report.json` for failures
 - Re-run import: `python UnrealStarter/BasketballGame/EditorPython/fel_import_ai_assets.py`
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Emergency Procedures

Service Recovery

```
# Restart all services
./scripts/stop_services.sh
./scripts/start_services.sh

# Docker recovery
docker-compose down
docker-compose up -d

# Check health
curl http://localhost:8888/healthz
```

Rollback

```
# Git rollback to last known good state
git log --oneline -10
git checkout <commit_hash>

# Rebuild
npm run build # frontend
./scripts/start_services.sh
```